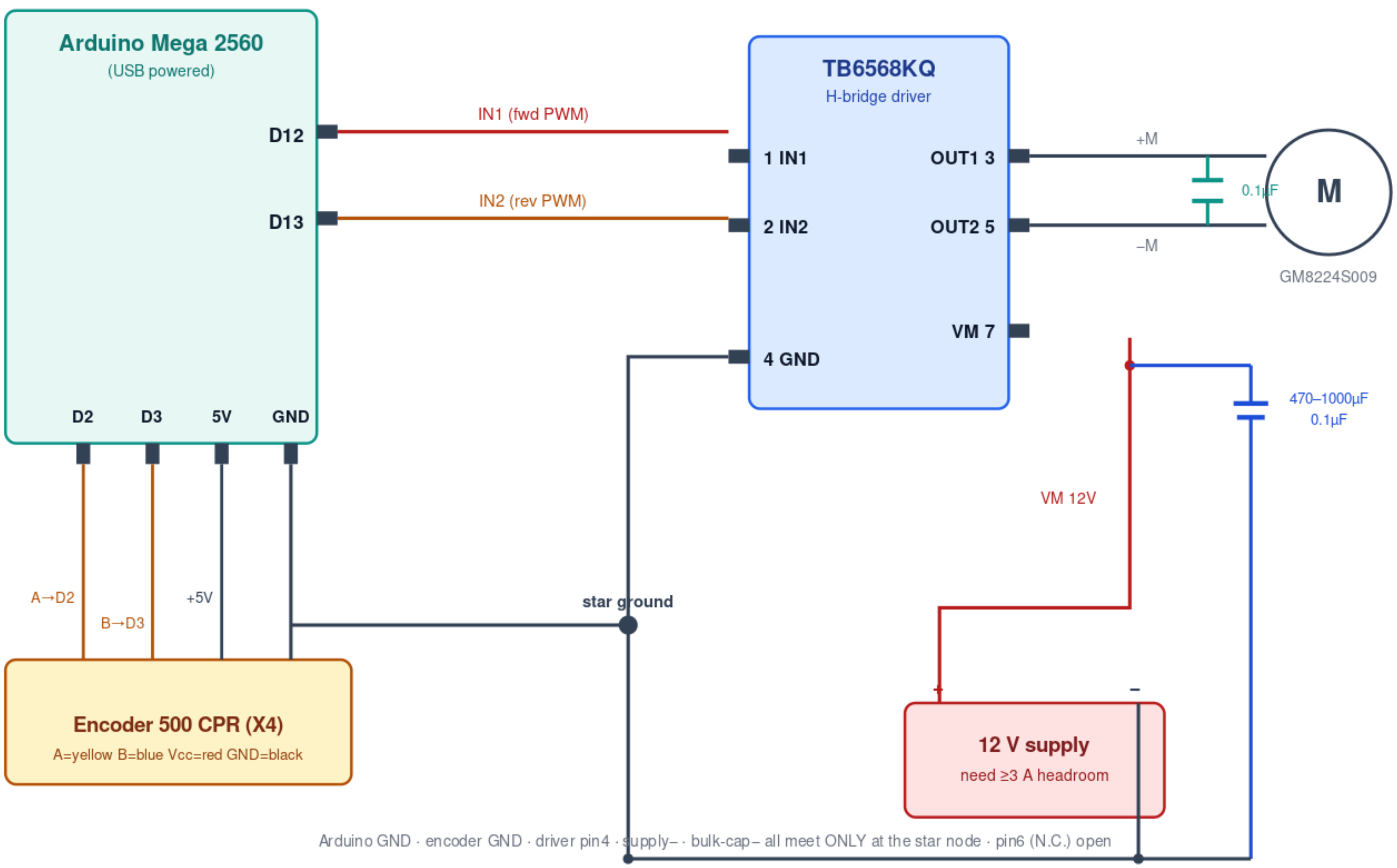
MSE 312 — Motor-Drive Circuits: Direct Drive vs 3-NAND

Two ways to connect the Arduino Mega 2560 to the TB6568KQ H-bridge. Both include the recommended power-integrity fixes (star ground, 12 V bulk cap, motor snubber).

Direct-drive wiring: Arduino Mega 2560 → TB6568KQ (no logic IC)

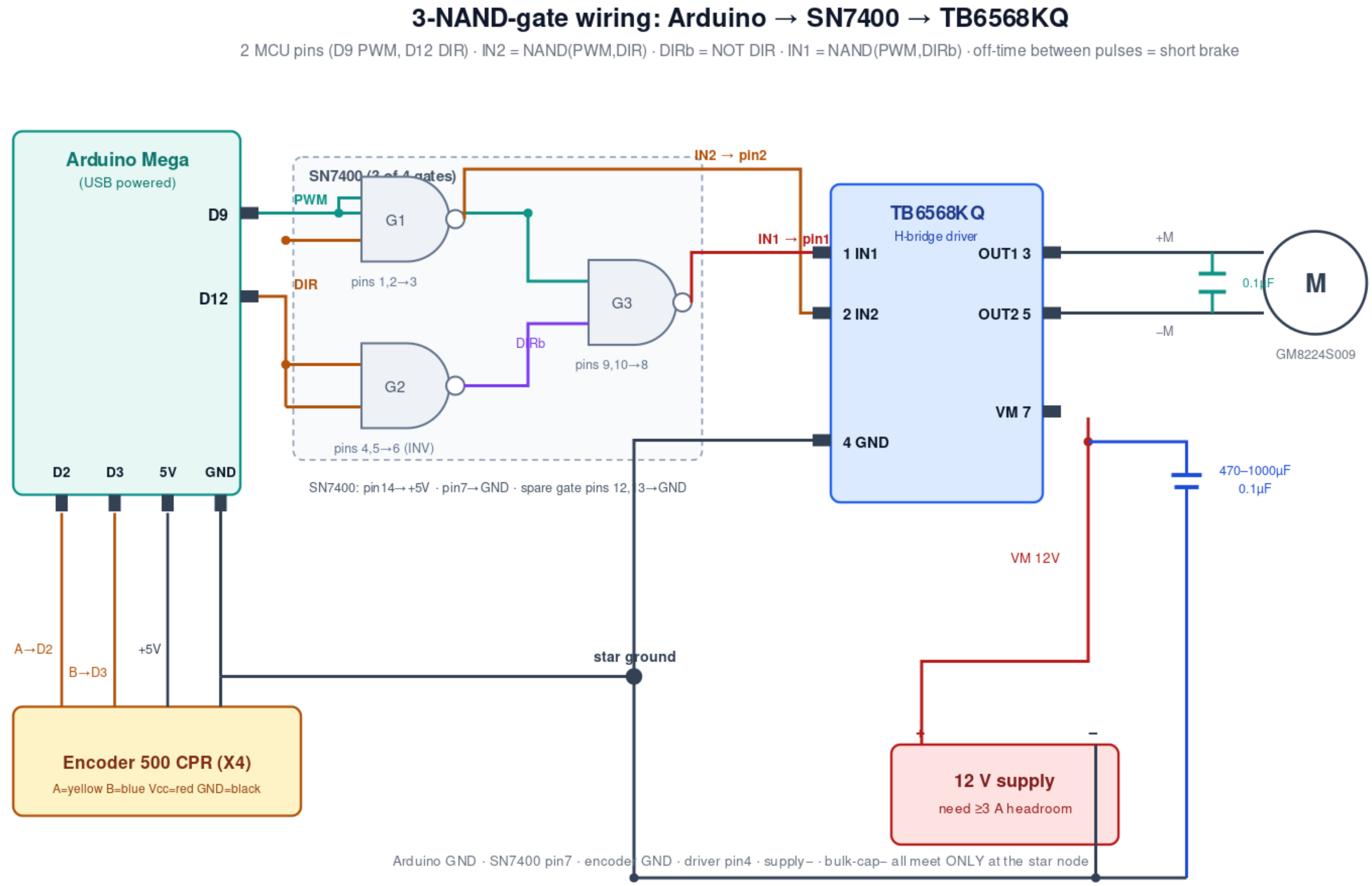
D12/D13 drive IN1/IN2 directly · PWM on the active direction pin · fwd=PWM on D12, rev=PWM on D13, brake=both HIGH, coast=both LOW



Direct drive (current build): D12→IN1, D13→IN2. Firmware PWMs the active direction pin (fwd = PWM on D12, D13 LOW; rev = PWM on D13, D12 LOW; brake = both HIGH; coast = both LOW). No logic IC. Note D12 is on Timer1 (~490 Hz) and D13 on Timer0 (~976 Hz), so fwd/rev PWM run at different low frequencies.

3-NAND-gate configuration

Uses one SN7400 (3 of 4 gates) to derive IN1/IN2 from just 2 MCU pins.



3-NAND drive: D9=PWM, D12=DIR. IN2 = NAND(PWM,DIR); DIRb = NOT DIR (gate as inverter); IN1 = NAND(PWM,DIRb). Gives sign-magnitude drive with short-brake between PWM pulses (slow decay). Two pins instead of three; keeps PWM on the single D9/Timer pin.